Club XYZ

Cloud Migration and Security Review

Cedric Titus

Cloud Engineer, B Cloud Engineers

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The Business Problem

Why Club XYZ needed better cybersecurity

1. **Outdated equipment**
   The older system needed modernization.

2. **Member information at risk**
   Personal information needed stronger protection.

3. **Limited monitoring**
   Suspicious activity was difficult to spot.

4. **No tested recovery plan**
   The club needed a reliable way to recover after a failure.

The Five-Week Assignment

What the project needed to accomplish

September 1 · October 1, 2026

1. **Move the System**
   Move the older member system to Microsoft Azure.

2. **Protect Member Information**
   Improve security and control who has access.

3. **Monitor Activity**
   Watch for suspicious activity and security problems.

4. **Test Recovery**
   Make sure the database can be restored after a failure.

5. **Review Costs**
   Explain the cost of running the new system.

Before and After

How the Azure project improved Club XYZ’s member system

Before

OLD SYSTEM

- Database was more exposed
- Passwords were stored
- Limited security monitoring
- No tested recovery plan

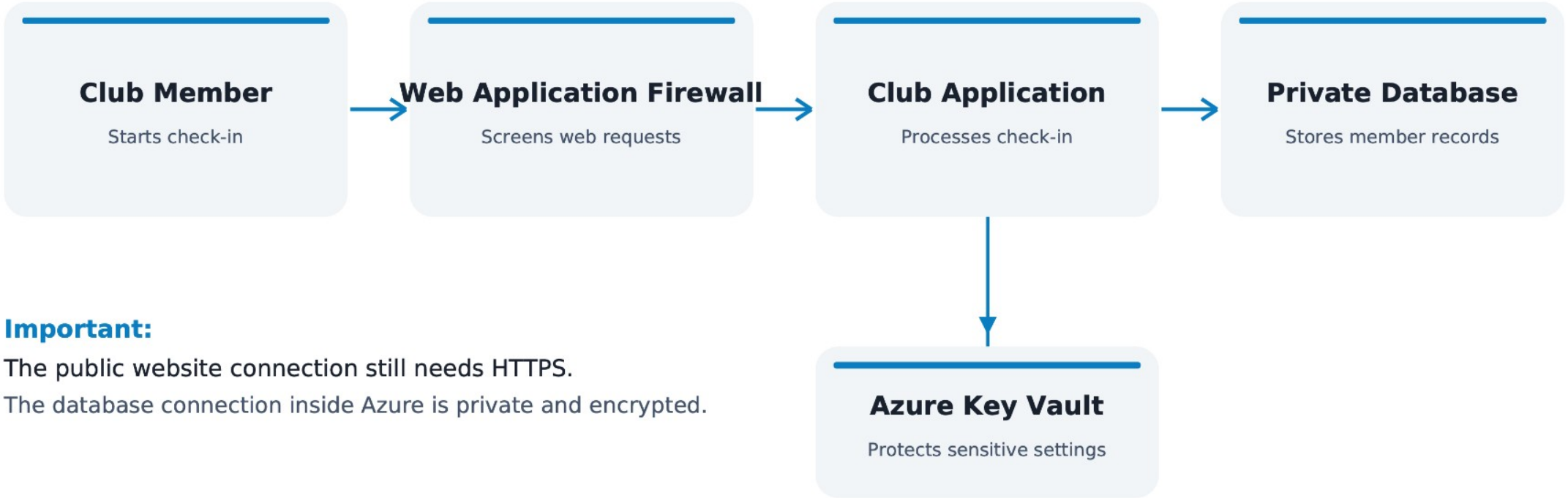
After

AZURE SYSTEM

- Private database connection
- Application has its own identity
- Firewalls, logs, and alerts
- Database recovery tested

How the System Works

How a member request travels through the Azure system



Important:

The public website connection still needs HTTPS.

The database connection inside Azure is private and encrypted.

Protecting Member Information

Multiple layers of encryption keep member data safe.

Microsoft Azure

Transport Layer Security (TLS)

Encrypts data while it's moving between the application and the database.



Transparent Data Encryption (TDE)

Scrambles member records while they're stored in the database.



Secret Store in Key Vault

Stores sensitive settings like connection strings in Azure Key Vault.



Public Website Encryption
The public website connection still needs HTTPS.

Additional Encryption Cost
$0 per month

An Identity for the Application

The application uses its own Azure identity instead of a saved database password.

Microsoft Azure

Microsoft Entra ID

Provides and manages the application's identity in Azure.

Managed Identity

Gives the application its own secure identity, without storing a database password.

Role-Based Access Control (RBAC)

Limits what the application can access. Permissions still need reviewing.

Additional Managed Identity Fee
$0 per month

Network Isolation, a private path to the database.

The database is not on the public internet.

Microsoft Azure

Virtual Network (VNet)

Creates a private network in Azure to isolate the application and database from the public internet.



Network Security Group (NSG)

Network Security Groups control which traffic is allowed or denied.



Private Endpoint (proposed, not yet implemented)

Azure Private Endpoint would give our database a private address for authorized app access. Public access to the database has been disabled.



Estimated Private Endpoint + DNS Cost

$7.82 per month

Inspection and Detection

How we detect suspicious activity and protect Club XYZ.

Microsoft Azure

Web Application Firewall (WAF)

Screens web traffic, blocks malicious requests, and helps protect the application from common attacks.



Log Analytics and Monitoring

Collects and stores security logs so suspicious activity can be investigated and trends can be identified.



Security Alerts

Alert rules notify us when suspicious activity is detected (e.g., blocked requests). Email delivery still needs verification.



Estimated Monthly Costs

Application Gateway (WAF)	**$273.31**
Log Analytics (Monitoring)	**$4.95**

Evidence from the Security Logs

Real security log evidence showing blocked requests and monitoring activity.

Microsoft Azure

Blocked attack patterns

772 blocked requests

September 24th through 29th, 2026

Blocking pattern	Count
Bot reputation	342
Attack-score threshold	277
False bot identity	153

Log Analytics Query (ApplicationGatewayFirewallLog)

Run | Time range : **Set in query** | Show : **Max.**

```
AzureDiagnostics
| where TimeGenerated >= datetime(2026-09-24T00:00:00Z)
|   and TimeGenerated < datetime(2026-09-29T18:00:00Z)
| where Category == "ApplicationGatewayFirewallLog"
| summarize
|    LogEvents = count(),
|    BlockedEvents = countif(action_s == "Blocked")
|    by _ResourceId, transactionId_g
| summarize
```

Results | Chart

FirewallLogEvents	BlockedLogEvents	RequestsWithFirewallLogs	BlockedRequests	LogEventsMissingTransactionID
3170	772	1428	772	0

24-hour snapshot

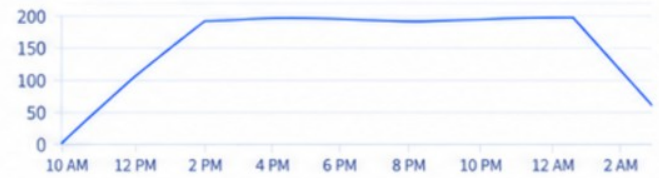
11 blocked requests

Dashboard's displayed 24-hour window

BCE Club Security Operations Dashboard

Application traffic and WAF activity from the last 24 hours, plus essential configuration verles from the last 7 days. Blocked requests are counted once per gateway hostname. The message chart counts log log entries, so one request can appear more than once.

Traffic over time — last 24 hours

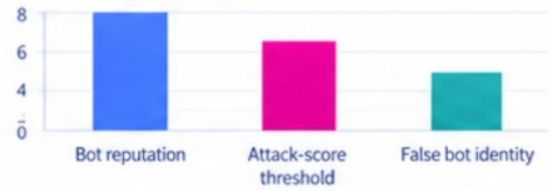


Blocked requests — last 24 hours

Unique blocked requests

11

WAF block messages — log entries, last 24 hours



Top source IPs — blocked requests, last 24 hours

Client IPs	Attempt(s)
97.74.230.245	7
185.214.82.52	2
45.112.186.164	2

Microsoft Defender for Cloud Secure Score

Microsoft Defender for Cloud Secure Score was not populated during the September 24th through 29th reporting period. Security recommendations and monitoring were enabled.

One Member’s Journey

Follow one test signup through the recorded security evidence.

TRACE-VIP-0927 | September 27, 2026 at 8:26 p.m. Eastern (September 28, 00:26 UTC)

01

Gateway & Firewall

Gateway and firewall inspected the request and allowed the legitimate signup.

00:26:24.000 UTC

→

02

Club Application

Processed the request to create a new member.

00:26:24.533 UTC

→

03

Azure SQL Database

Recorded a one-row operation and completed transaction.

Commit: 00:26:24.504 UTC

→

04

Member Dashboard

Displayed the new test member after the signup.

Creation minute: 00:26 UTC

Azure Key Vault securely provided the database connection setting before member signup.

Monthly Operating Cost

Estimated Azure infrastructure costs for a live-style Club XYZ setup.

Gateway & Web Application Firewall

$273.31

per month

Screens web requests and blocks common attacks.

All Other Azure Services

$38.79

per month

Application, database, monitoring and other modeled resources.

Monthly Infrastructure Total

$312.10

per month

$3,745.20 per year
at the same monthly estimate.

Largest cost: The gateway and firewall account for about 88% of the $312.10 monthly estimate.

Recommendation: Keep the firewall running continuously for a live club because it blocks malicious traffic and lowers risk.

Alternative option: Removing the gateway and firewall could save an estimated $273.31 per month, but would remove that layer of protection.

Estimate, not a bill.

Delivery Cost and Three-Year Plan

One-time delivery, ongoing Azure services, and routine support.



One-Time Delivery

$2,600

40 hours × $65 per hour

Azure Infrastructure

$312.10 / month

$11,235.60 over 36 months

Routine Support

$130 / month

2 hours × $65 per hour

Estimated Three-Year Total **$18,515.60**

Flexible platform, so the club can add tools as needs change.

Planning estimate, not a bill. Actual usage, rates, and support needs may change.

If Something Goes Wrong

We tested database recovery and measured how long it took.

6 min 26 sec

Database restore

Measured time to restore the database.

17–18 min

Restore + verification

Time to restore and confirm the data.

7 records

Data verification

Verified seven user records after recovery.

Proposed recovery targets — not yet guaranteed

Recovery Time Objective: 1 hour | Recovery Point Objective: 15 minutes

Measured lab restore is evidence; business recovery targets require approval and further testing.

What Still Needs Finishing

Security improvements to complete before the live club launch.

Public Website Encryption

Finish HTTPS for member traffic.
Cost: $130–$260 + certificate/domain

Login & Access Controls

Log sign-ins and enforce roles.
Cost: $195–$390 + log usage

Private Support Access

Build a private database support route.
Cost: $130–$260 + services

Alert Email Verification

Fix and test alert notifications.
Cost: $65–$130

Network Visibility

Enable network flow records.
Cost: $65–$130 + usage

Identity Sign-In Evidence

Confirm license and export sign-in logs.
Cost: $65–$130 + license/usage

Planning amounts: $650–$1,300 before listed extras. Scope and billing period to be confirmed.

Why the Security Role Continues

Keeping Club XYZ secure requires regular attention after migration.

Review Access

Check who can access member records and remove unnecessary permissions.

Investigate Alerts

Review suspicious activity and confirm security notifications work.

Maintain & Recover

Keep operating instructions current and regularly test database recovery.

Control Costs & Changes

Monitor Azure spending and review new tools before they are added.

Planning allowance: 2 hours of routine support per month at $65/hour = $130/month.

This is a support estimate, not full-time or 24/7 security coverage.

Recommendations and Next Steps

Priorities for preparing Club XYZ's member system for live use.

1 Secure the Public Website

Complete HTTPS and test the public connection.
Labor: $130–$260, plus certificate/domain costs.

2 Strengthen Sign-In Controls

Capture login records and review access.
Planning cost: $195–$390 + log usage.

3 Enable Private Support Access

Set up and test private database support.
Planning cost: $130–$260 + services.

4 Maintain Security Operations

Keep the firewall, test alerts, review access.
Alert testing: $65–$130; Azure: $312.10/mo.

Recommended order: HTTPS first → login checks → private support → ongoing reviews

Club XYZ — Final Business Summary

A more secure member system, with clear costs and a plan for what comes next.

B CLOUD ENGINEERS

COST

$312.10/month for Azure infrastructure.
$18,515.60 estimated over three years, including delivery and routine support.

KEY RISKS

Cyberattacks, data exposure, service downtime, and unfinished public website HTTPS.

RISK REDUCTION

Firewall screening, database encryption, security monitoring, and tested database restoration.

RECOMMENDATION

Keep the firewall for live use. Complete HTTPS and the remaining security checks before launch.

Thank you. I'm happy to take your questions.

Estimates, not bills. Production readiness requires completing and testing open items.